Concern Deforms a Learned Metric, But Its Finite-Capacity Exponent Is Effectively One-Dimensional

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Research-Derived Experiments · Paper B

Abstract

A goal or concern signal should not merely label a representation; it should reshape the metric by allocating resolution where errors matter. This paper reports two results. First, the original moved-location experiment remains a causal proof-of-concept: a reward at A or B locally increases the induced metric at the rewarded location, control-subtracted specificity is positive (+0.65 and +1.27), and larger-arena OOD decoding falls as local resolution is purchased. Second, a new 576-network Modal H100 sweep tests a derived rate-distortion exponent. The finite-capacity mechanism is confirmed, but the clean 2-D law is falsified as stated: at A=6, an anisotropic 2-D reward gives α=+0.309 [0.304,0.314], a stripe gives α=+0.302 [0.298,0.307], and a point reward gives α=+0.283 [0.278,0.288]. All SEs are <0.003, far below the preregistered 0.02 target. The honest conclusion is that concern deforms the metric, but this RNN/grid harness reallocates capacity with measured effective dimension d_eff≈0.8–1.0, not the normative 2-D exponent.

1. What Is Being Tested

The induced metric of a code r(x) is the pullback metric J(x)^T J(x): it says how much the population vector changes per unit movement in stimulus space. High metric density means fine local resolution. A reward-weighted path-integration objective asks the network to reduce errors near valuable locations. The causal question is whether that independent reward signal warps the learned metric.

The stronger theoretical question is quantitative. A high-resolution rate-distortion derivation predicts sqrt(det g) ∝ w^{d/(d+2)} under a finite capacity constraint. In a 2-D arena this gives α=1/2; in an effectively 1-D allocation it gives α=1/3. The Modal sweep was preregistered to decide between those possibilities.

2. Causal Metric Deformation

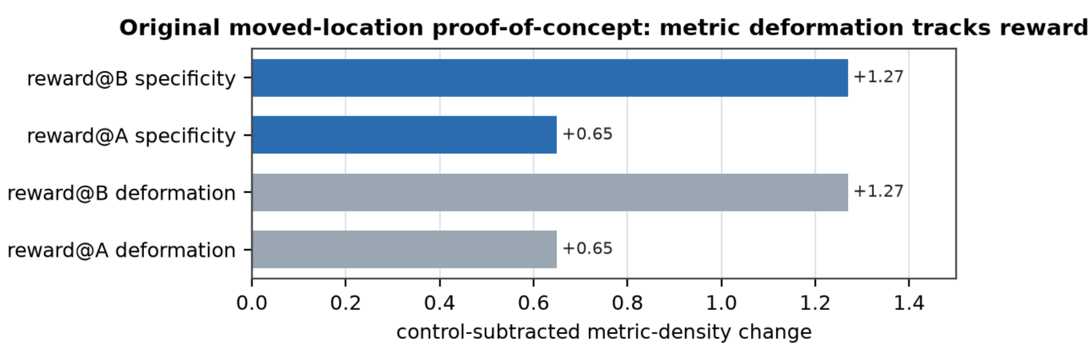

Figure 1. The original moved-location experiment (n=3 seeds) is the non-circular anchor: moving the reward moves the metric deformation. This specific A/B result was not rerun in the Modal exponent sweep.

The proof-of-concept remains important because the reward location is an independent variable, not a post-hoc label extracted from the representation. Reward@A raises metric density at A more than B; reward@B reverses the asymmetry. A no-reward control is flat. This establishes that concern can deform the learned metric, while leaving open how that deformation scales under capacity.

3. Modal Exponent Sweep

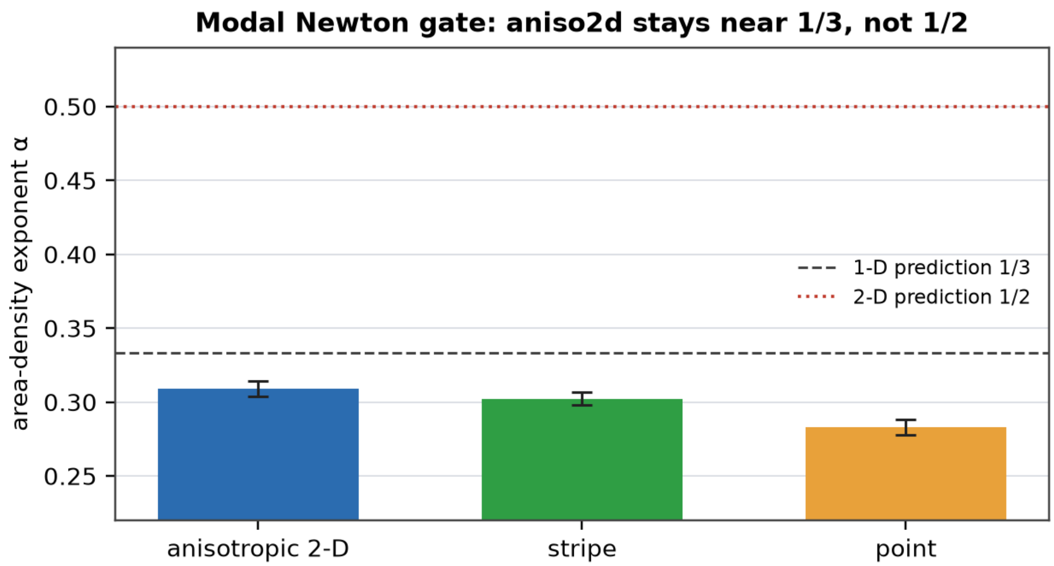

Figure 2. At the primary amplitude A=6, all geometries stay near the 1-D exponent family. The 2-D prediction α=1/2 is decisively excluded.

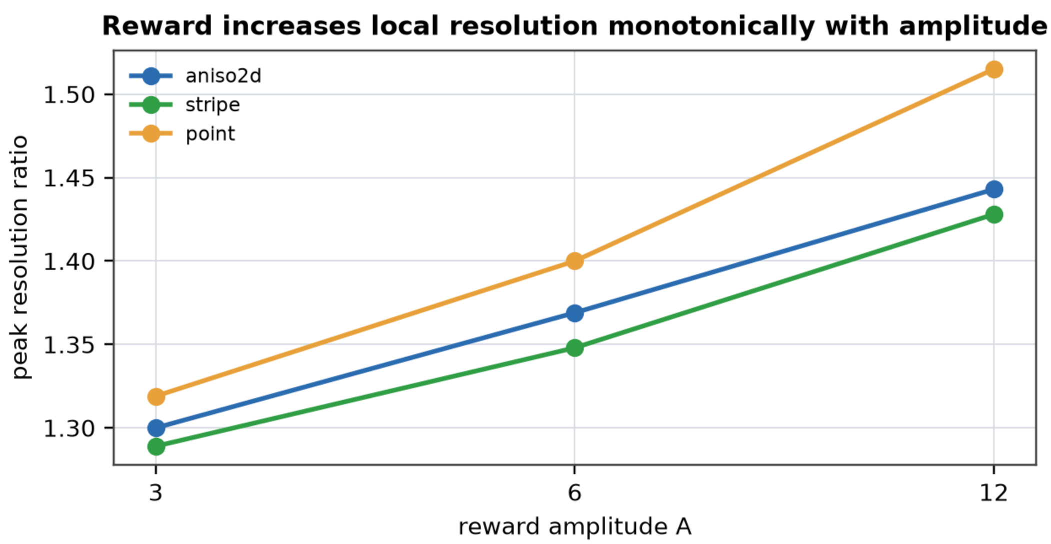

Figure 3. Peak resolution increases monotonically with reward amplitude, confirming value-driven reallocation even though the exponent is not 2-D.

Geometry	α at A=6	95% CI	SE	d_eff	R^2
aniso2d	+0.309	[0.304, 0.314]	0.0025	0.90	0.528
stripe	+0.302	[0.298, 0.307]	0.0023	0.87	0.565
point	+0.283	[0.278, 0.288]	0.0025	0.79	0.417

Table 1. Primary preregistered exponent gate. Aniso2d does not separate from stripe and is much closer to 1/3 than 1/2.

4. Discussion

The right conclusion is a measured law, not the desired one. Capacity is load-bearing: without it the exponent was near +0.07; with it, reward density reliably predicts metric density with α around +0.3. But the code does not use a full two-dimensional allocation even when the value field is two-dimensional. The architecture appears to move resolution along an effectively one-dimensional bottleneck or gradient-like degree of freedom.

This result is scientifically useful because it narrows the phenomenon. The paper should not claim a confirmed 2-D Newton law. It should claim: a concern signal causally warps a learned metric; a finite-capacity bottleneck turns that warp into a stable power law; and the measured exponent reveals the effective allocation dimension of the learned code.

5. Limitations

The moved-location A/B specificity result is still n=3. The 576-network Modal run resolves the exponent and amplitude question, not the moved-location design. The reward is a loss weight, not a full reinforcement-learning value head. The derivation remains a normative high-resolution efficient-coding law; this paper measures how a particular RNN/grid harness departs from it.

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